

BUY, Russian Federation

WMO: 272420

Lat: 58.483N

Lon: 41.533E

Elev: 103

StdP: 100.09

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-25.5	-31.7	0.2	-28.8	-28.2	0.3	-25.4	6.0	-8.5	5.3	-6.9	0.8	30	0.545

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	29.6	20.6	27.4	19.8	25.5	18.8	21.9	27.5	20.7	25.9	19.7	24.2	1.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.9	14.8	24.6	18.9	13.8	23.7	17.8	13.0	22.4	64.2	27.4	60.3	25.9	56.6	24.3	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
5.3	4.5	4.0	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-32.6	32.0	3.7	2.4	-35.3	33.7	-37.4	35.1	-39.5	36.4	-42.2	38.1		
			-32.8	23.4	3.7	1.2	-35.4	24.3	-37.5	25.0	-39.6	25.7	-42.2	26.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.1	-10.3	-9.1	-3.6	4.6	11.4	15.9	19.0	15.9	10.4	4.1	-2.7	-7.3
	DBStd	11.41	7.71	7.48	4.99	4.89	4.99	4.04	3.59	3.71	4.00	4.39	5.67	7.51
	HDD10.0	2951	628	536	421	173	43	3	0	2	43	188	379	536
	HDD18.3	5292	886	769	679	411	219	94	36	94	239	441	629	794
	CDD10.0	804	0	0	0	12	88	181	278	185	54	6	0	0
	CDD18.3	103	0	0	0	0	6	22	55	19	1	0	0	0
	CDH23.3	977	0	0	0	3	66	230	502	168	7	0	0	0
	CDH26.7	261	0	0	0	0	16	60	137	47	1	0	0	0
Wind	WSAvg	1.7	1.9	1.9	1.9	1.7	1.8	1.5	1.3	1.4	1.5	1.8	1.9	1.9
Precipitation	PrecAvg	615	43	35	35	33	47	71	74	67	54	63	50	44
	PrecMax	842	76	68	71	73	130	137	168	208	106	134	95	77
	PrecMin	465	20	5	3	15	12	17	1	8	22	23	10	17
	PrecStd	96	14	18	18	16	26	33	45	43	22	31	22	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	3.1	10.7	22.7	29.1	31.1	32.7	31.1	24.1	17.6	9.5	5.9
		MCWB	1.5	1.5	5.3	13.6	19.3	21.2	21.9	21.5	17.6	13.0	8.6	5.3
	2%	DB	1.2	1.8	6.6	19.1	25.1	28.5	29.8	27.2	21.1	14.0	6.9	3.5
		MCWB	0.6	1.0	2.9	11.7	17.2	19.8	20.8	19.9	16.2	11.0	6.0	2.7
	5%	DB	0.6	1.1	4.4	15.9	22.5	26.1	27.9	24.9	18.7	11.7	5.3	1.8
		MCWB	0.1	0.5	1.7	10.0	14.7	18.8	20.3	18.7	14.6	9.6	4.6	1.2
	10%	DB	-0.2	0.4	2.9	13.0	20.4	23.8	26.1	22.8	16.7	10.1	3.8	0.9
		MCWB	-0.7	-0.2	1.1	7.9	13.4	17.4	19.5	17.8	13.6	8.6	3.1	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	2.0	5.9	14.3	21.0	22.8	23.8	23.1	19.5	13.6	8.9	5.3
		MCDB	2.0	2.6	9.7	21.3	27.5	28.7	29.4	29.1	22.2	16.5	9.7	5.8
	2%	WB	0.8	1.2	3.5	12.3	18.1	20.9	22.2	21.0	17.0	11.4	6.2	2.8
		MCDB	1.1	1.6	5.6	18.3	23.2	26.9	27.9	25.6	19.8	13.3	6.6	3.2
	5%	WB	0.2	0.6	2.2	10.4	16.1	19.6	21.1	19.6	15.3	9.9	4.7	1.3
		MCDB	0.6	1.0	3.8	15.2	20.9	24.6	26.4	23.7	17.9	11.2	5.2	1.6
	10%	WB	-0.7	-0.2	1.3	8.5	14.4	18.2	20.1	18.2	13.7	8.7	3.1	0.5
		MCDB	-0.2	0.4	2.6	12.5	18.9	22.5	24.7	22.1	16.3	10.0	3.6	0.9
Mean Daily Temperature Range	5% DB	MDBR	5.9	7.3	8.7	9.8	10.8	10.3	10.4	9.5	8.2	5.4	4.0	4.8
		MCDBR	4.1	4.9	10.2	14.4	13.8	13.5	12.9	12.9	11.4	8.4	4.8	3.2
		MCWBR	4.1	4.5	6.9	8.2	6.6	6.0	5.5	6.3	6.7	5.6	4.4	3.3
	5% WB	MCDBR	4.0	4.6	7.6	13.5	11.8	11.8	11.4	11.4	9.8	7.5	4.4	3.2
		MCWBR	4.0	4.5	5.4	7.9	6.7	5.9	5.7	6.0	6.2	5.4	4.3	3.3
Clear-Sky Solar Irradiance	taub	0.278	0.316	0.345	0.396	0.372	0.367	0.404	0.400	0.363	0.342	0.306	0.266	
	taud	2.066	2.056	2.058	2.132	2.378	2.410	2.329	2.317	2.404	2.345	2.178	2.044	
	Ebn at Noon	520	677	775	802	860	869	822	795	771	665	489	401	
	Edh at Noon	53	87	116	129	108	107	113	107	84	67	48	39	
All-Sky Solar Radiation	RadAvg	0.41	1.16	2.58	3.94	5.27	5.52	5.45	4.18	2.42	1.05	0.36	0.22	
	RadStd	0.07	0.23	0.28	0.44	0.41	0.46	0.55	0.29	0.31	0.18	0.06	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.80	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.66	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page